

Final Exam: Micro 629 - Fall 2012

Texas A&M University

December 11th, 2012

By writing down your name you pledge to obey TAMU Honor Code:

“An Aggie does not lie, cheat, or steal or tolerate those who do.”

Name (in pen, not pencil): _____

EXAM RULES

- Your interpretation of the questions makes part of the exam. Thus, no questions will be answered during the exam.
- No personal items (bags, cellphone, books, notebooks, calculators, etc.) will be allowed with you during the exam.
- Some of the sections in the different problems require you to prove the preceding sections in the problem; if you cannot prove one, move on, and assume that it is true; use it to prove what is next and then come back if you have time.
- It is valid to use results from class or homework, as long as: (i) I am not asking you to provide an explicit proof, and (ii) you precisely quote the result from class or homework. For instance, it is not valid to write: “by a result from class the proof is completed.” In this case you would have to quote the exact result, with all hypotheses and implications.

		grade
1	(a)	
	(b)	
	(c)	
2	(a)	
	(b)	
	(c)	
	(d)	
3	(a)	
	(b)	
	(c)	
	(d)	
4	(a)	
	(b)	
	(c)	
Total		

1. (20/100) Let Z be a finite set of prizes and $\succsim$ be a preference over the space of lotteries $\Delta(Z)$.
 - (a) (5/20) State the independence axiom (IA) for $\succsim$.
 - (b) (7.5/20) Assume that $\succsim$ is complete and transitive and satisfies IA. Prove that for each triple $\{\pi, \pi', \pi''\} \subseteq \Delta(Z)$ and each $\alpha \in (0, 1)$, $\pi \succ \pi'$ if and only if $\alpha\pi + (1 - \alpha)\pi'' \succ \pi' + (1 - \alpha)\pi''$.
 - (c) (7.5/20) Assume that $\succsim$ is complete and transitive and satisfies IA. Prove that for each 4-tuple $\{\pi, \pi', \pi'', \pi'''\} \subseteq \Delta(Z)$ such that $\pi \succ \pi'$ and $\pi'' \succ \pi'''$ and each $\alpha \in [0, 1]$, we have that $\alpha\pi + (1 - \alpha)\pi'' \succ \pi' + (1 - \alpha)\pi'''$.
2. (40/100) Consider an agent who has wealth w and is an expected utility maximizer with utility index over amounts of money u . There are n possible states of the world. State $i \in \{1, \dots, n\}$ is realized with probability α_i . There are n assets the agent may buy: one unit of asset $i \in \{1, \dots, n\}$ returns one unit of money in state i and zero in each state $j \neq i$. The price of asset i is $p_i > 0$. Denote by z_i the amount of asset i the agent buys.
 - (a) (7.5/40) Write down the agent's optimal portfolio choice problem (utility maximization problem).
 - (b) (7.5/40) Suppose that there are only two states $\{1, 2\}$, the agent's utility index is $u(z) \equiv \log(z)$, and the agent has bought a portfolio (z_1, z_2) . Before the state of the world is realized, the agent is approached by another agent who offers to buy (z_1, z_2) from her in exchange for a payment c that is independent of the state of the world (a risk-free asset). What is the minimum c , as a function of (α_1, α_2) and (z_1, z_2) , for which the agent is willing to accept the deal?
 - (c) (10/40) Suppose that there are only two states $\{1, 2\}$ and there are K agents with identical utility indices $u_k(z) \equiv \log(z)$ for $k = 1, \dots, K$. Agent k has initial wealth w_k . Prove that the aggregate optimal portfolio is a function of prices and aggregate wealth $w \equiv w_1 + \dots + w_K$.
 - (d) (15/40) Suppose that there are n states and there are k agents with identical utility indices $u_k(z)$ that are homogeneous of degree $r > 0$ and non-negative. Agent k has initial wealth w_k . Prove that the aggregate optimal portfolio is a function of prices and aggregate wealth $w \equiv w_1 + \dots + w_k$.
3. (20/100) Consider a one-input one-output production model. Let $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ be an increasing production function.
 - (a) (5/20) What is the production set associated with f , Y^f ?
 - (b) (5/20) Assume that the price of the input is fixed. Let $c^f(q)$ be the cost of producing q units for the firm. Prove that if f is concave, then the cost function is convex.
 - (c) (5/20) Define Non-increasing (NI) returns to scale of a production set Y .
 - (d) (5/20) Assume now that Y^f satisfies NI returns to scale. Is c^f convex? (provide a formal argument that supports your answer).
4. (20/100) Let $Y \subseteq \mathbb{R}^L$ be a production set.
 - (a) (5/20) State the definition of additivity of a production set.
 - (b) (5/20) Argue in one paragraph that the production set of a firm in an industry in which there is free entry is always additive.

- (c) (10/20) A private ownership economy is a collection of agents N represented by preferences $(\succsim_i)_{i \in N}$ and firms $J = \{1, \dots, J\}$ represented by production sets $(Y^j)_{j \in J}$. Agent $i \in N$ has initial endowment of resources $w^i \in \mathbb{R}_+^L$ and for each firm $j \in J$, agent i owns a proportion θ_{ij} of firm j . A competitive equilibrium for the economy is a triple $(x \equiv (x^i)_{i \in N}, y \equiv (y^j)_{j \in J}, p \equiv (p_l)_{l=1}^L)$, where $x^i \in \mathbb{R}_+^L$ is agent i 's consumption, $y^j \in \mathbb{R}^L$ is firm j 's production vector, and p is a price vector, that satisfy among other conditions, that for each firm $j \in J$, y^j maximizes firm j 's profits at prices p . Prove that if firm j belongs to an industry in which there is free entry and possibility of inaction, then in any competitive equilibrium firm j 's profit is zero.